ARTICLE

Bidirectional Link Between Exercise and the Gut Microbiota

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HUANG, B., L. ZHAO, and S.C. CAMPBELL. Bidirectional link between exercise and the gut microbiota. *Exerc. Sport Sci. Rev.*, Vol. 52, No. 4, pp. 132–144, 2024. *Exercise is well known to exert beneficial changes to the gut microbiota. An emerging area is how the gut microbiota may regulate exercise tolerance. This review will summarize the current evidence on how exercise influences gut microbial communities, with emphasis on how disruptions or depletion of an intact gut microbiota impacts exercise tolerance as well as future directions.* **Key Words:** short-chain fatty acids, exercise, skeletal muscle, germ-free, gnotobiotic, antibiotics

KEY POINTS

- Exercise exerts beneficial effects on the gut microbiota.
- Reducing or eliminating the gut microbiota reduces exercise capacity.
- Rationale for the reduction in exercise can be due to the alteration in the gut microbiota–skeletal muscle axis.

INTRODUCTION

The gut microbiota comprises trillions of microbial cells, including bacteria, fungi, viruses, and archaea (1). Particularly for bacteria, there are over 1100 genera in the gut, and approximately 90% of gut bacteria fall under the phyla Bacteroidota and Bacillota (formerly known as Bacteroidetes and Firmicutes (2)), respectively. Meanwhile, the minority of gut bacteria are the Pseudomonadota, Actinomycetota, Fusobacteriota, and Verrucomicrobiota (formerly known as Proteobacteria, Actinobacteria, Fusobacteria, Verrucomicrobia (2), respectively) phyla (3). Commonly observed in a healthy gut microbiota is a decreased Bacillota to Bacteroidota ratio, stable community, and greater species diversity (4).

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The gut microbiota plays a significant role in human health; when the gut microbiota is in symbiosis, microbes can promote health; however, when in dysbiosis (unbalanced gut microbes), microbes contribute to chronic disease. In healthy circumstances, the gut microbiota aids in digestion, nutrient absorption, and production of folate, vitamins, and short-chain fatty acids (SCFAs) (5).

Several groups, including our lab (4), have examined the link between the gut microbiota and exercise in animal models (6–8). Exercise not only is central to healthful aging but has also been shown to improve the diversity of microbes within the Bacillota phylum (4,8,9), increasing the abundance of beneficial bacteria such as *Roseburia intestinalis*, *Faecalibacterium prausnitzii*, and *Akkermansia muciniphila* (10,11). Although the overall influence of exercise on the gut microbiome is becoming more defined, the reverse knowledge is lacking (*i.e.*, what is the impact of microbiota depletion on enhanced exercise performance following exercise training and the mechanisms mediating the enhanced exercise performance). Therefore, the purpose of this review paper is to analyze how exercise alters the gut microbiota composition and report the influence of the presence or absence of the gut microbiome on exercise performance (summary of all animal and human studies presented in this review can be found in Table 1).

Exercise-Induced Gut Microbiota Changes

Moderate and voluntary exercise versus forced exercise in animal models

Research studies use a variety of methods to exercise animals. Voluntary wheel running (VWR) and forced treadmill running (FTR) are the two most used. With VWR exercise, wheels are

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TABLE 1. Summary of all animal and human studies presented in this review.

Animal Models

Article	Subjects	Exercise Stimulus	Diet	Test Groups	Main Outcomes	Gut Microbiota Findings	Metabolites/Metabolic Pathways Findings
Campbell *et al.* (11)	*N* = 36 male mice	12 wk of free access to running wheel in exercise groups	*Ad libitum* normal vs high-fat diet (45% kcal from fat)	Lean sedentary Obese sedentary Lean exercise Obese exercise	↓ Intestinal inflammation (VWR) Exercise alters gut microbiota independent of diet	*Clostridiales* found in all groups Exercise-specific changes: *Faecalibacterium prausnitzi*, *Clostridium* spp., *Allobaculum* spp.	COX-2 expression OS >LS > LX > OX
Matsumoto *et al.* (12)	*N* = 14 male rats	5 wk of free access to running wheel in exercise groups	*Ad libitum* 25% casein-sucrose diet	Control (*n* = 7) VWR (*n* = 7)	↑ *n*-butyrate concentration and cecum size (VWR)	↑ Butyrate-producing bacteria	
Allen *et al.* (6)	*N* = 29 male rats	6 wk VWR: free access to running wheel for 30 d FTR: 40 min·d^{-1} at 8–12 m·min^{-1}, 5% grade, 5 d·wk^{-1}	*Ad libitum* commercial diet	Sedentary (*n* = 9) VWR (*n* = 10) FTR (*n* = 10)	VWR and FTR alter gut microbiome differently	↓ α-Diversity in VWR compared with FTR or SED ↓ *Turcibacter* spp. (VWR)	
Allen *et al.* (8)	Donor and GF mice	6 wk VWR: free access to running wheel	*Ad libitum* diet (22.37% protein, 15.6% fat, 61.98% carbohydrate and 4.4% of fiber)	Donor–voluntary exercise mice (*n* = 10) Donor–control mice (*n* = 9) Recipient–control (*n* = 15) Recipient–exercise (*n* = 16) Recipient–control–DSS (*n* = 16) Recipient–exercise–DSS (*n* = 16)	Recipient of exercise-trained donor caused attenuated response to colitis (reduced colon shortening and mucus depletion, increased tissue regenerating cytokine expression)	β-Diversity changed in recipient mice 5 wk after colonization from donor mice exercise changed β-diversity	↑ TGFβ, Fox P3, IL-22 in recipient–exercise–DSS
Lamoureux *et al.* (13)	*N* = 42 mice	55 d VE: free access to running wheel FE: 5 d·wk^{-1}, 40 min, 15–20 m·min^{-1}	Premeasured consistent diet	VE (*n* = 10) FE (*n* = 11) Sedentary (*n* = 21)	Similar inflammatory markers and β-diversity across all groups	Algorithm predicted novel taxa: *Rikenellaceae* and *Lachnospiraceae* to be associated with exercise	
Mika *et al.* (14)	*N* = 40 male rats	Exercise groups: 6 wk of free access to running wheel	*Ad libitum* diet of 18.6% protein, 6% fat and 3.5% fiber	Juvenile onset exercise (*n* = 10) Adult onset exercise (*n* = 10) Sedentary (*n* = 10 juvenile, *n* = 10 adult)	↑ Bacteroidota and ↓ Bacillota in juvenile exercise ↑ Genera altered in juvenile exercise Persistent increase in lean body mass in juvenile exercise absent in adult exercise	↓ Species richness in juvenile exercise compared to juvenile sedentary rats ↑ *Blautia* spp., *Anaerostipes* spp., *Methanosphaera* spp. in juvenile exercise ↑ *Rikenellaceae* spp. in adult exercise ↓ *Desulfovibrio* spp. in juvenile exercise	
Hsu *et al.* (7)	*N* = 24 male mice	Exhaustive swim test	*Ad libitum* standard chow diet	SPF (*n* = 8) GF (*n* = 8) B. *fragilis* (BF) gnotobiotic (*n* = 8)	↑ Swim time in SPF > BF > GF ↑ Liver, muscle, brown adipose, epididymal fat pad weight in SPF and BF		↑ Serum glutathione peroxidase and catalase in SPF ↓ Serum superoxide dismutase in BF
Scheiman *et al.* (15)	*N* = 32 mice	Treadmill run to exhaustion		*V. atypica* gavaged mice *L. bulgaricus* gavaged mice	↑ Run time of *V. atypica* gavage *V. atypica* metabolizes lactate into propionate ↑ Run time of mice administered with propionate		↓ Levels of inflammatory cytokines in *V. atypica* mice
Huang *et al.* (16)	Gnotobiotic mice/GF mice	Exhaustive swim test Swim training: 10 min·d^{-1}, 5 d·wk^{-1} for 5 wk	*Ad libitum* sterile diet	*E. rectale* inoculated *C. coccoides* inoculated *L. plantarum* TWK10 inoculated GF (*n* = 6–8/microbe)	↑ Basal metabolic rate in *C. coccoides* and *L. plantarum* Pathological hepatic necrosis in *C. coccoides* ↑ Swim time in *E. rectale* no performance improvement in *L. plantarum* and *C. coccoides*		↑ Serum creatine kinase in *C. coccoides* ↓ Lactate in *E. rectale* and *C. coccoides* ↑ Serum glucose in *E. rectale* ↓ Hepatic glycogen in SPF mice

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TABLE 1. *(Continued)*

Animal Models

Article	Subjects	Exercise Stimulus	Diet	Test Groups	Main Outcomes	Gut Microbiota Findings	Metabolites/Metabolic Pathways Findings
Valentino *et al.* (17)	*N* = 40 female mice	9 wk progressive weighted wheel running: 1 g added per week up to 6 g during weeks 7–9	*Ad libitum* irradiated chow	Untreated + nonrunning control (CU) ABX treated + nonrunning control (CT) Untreated + progressive weighted wheel running (PU) ABX treated + progressive weighted wheel running (PT) (*n* = 10–11/group)	↓ Hypertrophy of soleus and plantaris muscle fibers in PT than PU Blunted fiber type shift from 1 to >2a in PT compared to PU Gut microbiome helps skeletal muscle adaptation to exercise		No change in serum cytokines
Okamoto *et al.* (18)	Male mice	Treadmill run time test before sacrifice Part 1: 2 wk (ABX vs control) Part 2: 6 wk (LMC vs HMC) Part 3: 3wks (ABX+ saline vs ABX+ butyrate) Part 4: 6 wk + 3 d (LMC vs LMC + FMT + inulin)	Measured daily food intake of regular chow diet	Part 1: ABX treated Control (*n* = 6/group) Part 2: Low microbiome accessible carbohydrate diet (LMC) High microbiome accessible carbohydrate diet (HMC) Part 3: ABX + saline ABX + butyrate Part 4: LMC LMC + FMT + inulin	Acetate is an important energy substrate in endurance exercise ↓ Run time in 2 wk of ABX treatment Undetectable levels of SCFA (acetate, propionate, butyrate) in ABX treated ↓ Exercise capacity and SCFA in LMC compared to HMC Continuous acetate infusions restored exercise capacity ↑ Run time in LMC + FMT + inulin compared to LMC		
Dowden *et al.* (19)	Male mice	FTR: 70% max speed, 1 $h \cdot d^{-1}$, 5 $d \cdot wk^{-1}$, for 5 wk	*Ad libitum* irradiated standard chow	Examined preexercise training, postexercise training, and 1 wk after ABX	Exercise training increases run to exhaustion time ABX abolished run time improvements post-training		ABX abolished exercise training increases of hindlimb blood flow, AMP-activated protein kinase, sirtuin-1, PGC-1a citrate synthase, nitric oxide
Nay *et al.* (20)	Male mice	Treadmill run to exhaustion	*Ad libitum* diet of 61.3% protein carbohydrates, 25.2% protein, and 13.5% lipids	Control (*n* = 10) ABX for 21 d (*n* = 10) → Natural reseeding for 10 d (*n* = 9)	↓ Running endurance and extensor digitorum longus muscle fatigue index in ABX mice Natural reseeding normalized muscle endurance capacity		↑ Gene expression of SCFA chain, glucose transporters, sodium-glucose cotransporter 1, muscle glycogen levels after reseeding
Hutchinson *et al.* (21)	*N* = 56 male and female mice	6 wk VWR: free access to running wheel	Measured daily food intake of commercial diet	Control VWR ABX VWR + ABX GF VWR + GF	↓ VWR times in GF and ABX compared to control ↓ Treadmill endurance post-training in GF mice		↑ Citrate synthase enzyme activity in all VWR groups ↑ PGC1a expression in muscle of VWR
Dohnalova *et al.* (22)	Male and female mice	VWR: free access to running wheel Treadmill run test	*Ad libitum* normal chow diet	ABX Control ABX + exercise Control + exercise	Microbiome-dependent metabolites stimulate TRPV1-expressing neurons and dopamine during exercise Lack of gut microbiome blunted exercise-induced dopamine surge		
Lahiri *et al.* (23)	Male mice	Treadmill run to exhaustion test	*Ad libitum* autoclaved chow	GF Pathogen-free GF colonized by pathogen free gut microbiota GF + SCFAs Pathogen-free + penicillin	Atrophy in skeletal muscle of GF mice ↑ Muscle mass and oxidative metabolic capacity of muscles when GF inoculated with pathogen-free gut microbiota Skeletal muscle impairments reversed in GF + ABX		↓ Insulin-like growth factor 1 in GF mice ↓ Skeletal muscle growth and mitochondrial function gene transcription in GF mice ↓ Serum choline in GF mice

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Human Models

Article	Subjects	Exercise Stimulus	Test Groups	Main Outcomes	Gut Microbiota Findings	Metabolites/Metabolic Pathways Findings
Bressa *et al.* (10)	*N* = 40 women	Physical activity measured with accelerometer over 7-d period	Sedentary women: those that did not meet WHO exercise criteria Active women: 3+ h·wk^{-1} of exercise	Breaking of sedentary behavior induces gut microbiota changes	Exercise-specific changes: *F. prausnitzii*, *R. hominis*, *A. muciniphila*	Cysteine aminopeptidase higher in active group
Langsetmo *et al.* (24)	*N* = 373 community dwelling older men Average age: 84 yr	Activity assessed with SenseWear Pro3 Armband over 7-d period	Community-dwelling older men	Physical activity not associated with α-diversity	↑ *Cetobacterium* and ↓ *Coprobacillus*, *Adlercreutzia*, *Eryseipelotrichaceae* abundances associated with ↑ step count	
Bai *et al.* (25)	*N* = 267 children aged 7–18 yr		Data gathered from American Gut Project	Factors affecting gut microbiota: BMI, exercise frequency, diet	Actinomycetota phylum depleted OTUs, Pseudomonadota phylum enriched OTUs with respect to BMI level, exercise frequency, diet type Bacillota phylum enriched OTUs with higher exercise frequency	
Morita *et al.* (26)	*N* = 32 sedentary women 65 +yr	12 wk TM: 1 h group resistance training strengthening trunk muscles AE: 60 min daily brisk walk at ≥3 METs intensity	Trunk muscle training (TM) (*n* = 14) Aerobic exercise training (AE) (*n* = 18)	↑ *Bacteroides* associated with time spent briskly walking	↑ *Bacteroides* abundance in AE	
Munukka *et al.* (27)	*N* = 19 sedentary overweight women	6-wk endurance exercise intervention using bicycle ergometer of 85% of individually predicted heart rate max	Sedentary overweight women (BMI >27.5 kg·m^{-2})	↑ Akkermansia and ↓ Pseudomonadota with exercise ↓ Phospholipids and VLDL after exercise		↓ Fructose and amino acid metabolism without major systemic metabolic changes with training No change in CRP ↓ Proinflammatory activity of VAP-1 with training
Allen *et al.* (28)	*N* = 32 lean and obese sedentary individuals	6-wk supervised endurance exercise for 3 d·wk^{-1} progressing from 30-60 min·d^{-1} and 60%–75% heart rate reserve	Lean (*n* = 18) Obese (*n* = 14)	β-Diversity dependent on obesity status ↑ Fecal SCFA in exercising lean individuals only	Exercise-induced gut microbiota changes reversed once training ended	
Barton *et al.* (9)	*N* = 86		Professional rugby players (*n* = 40) Sedentary controls (*n* = 46)	↑ SCFA (acetate, propionate, butyrate) in athletes		↑ Amino acid and ABX biosynthesis pathways, carbohydrate metabolism in athletes
Liang *et al.* (29)	*N* = 28 professional martial arts athletes		Higher-level athletes (*n* = 12) Lower-level athletes (*n* = 16)	↑ Shannon and Simpson diversity index in higher-level athletes	↑ *Parabacteroides*, *Phascolarctobacterium*, *Oscillibacter*, *Bilophila* in higher-level athletes ↑ *Megasphaera* in lower-level athletes *Parabacteroides* positively correlated with exercise time	↑ Histidine and carbohydrate metabolism in higher-level athletes
Han *et al.* (30)	*N* = 19 Chinese female rowing athletes aged 12–26 yr Adult = 19+ yr		Adult elite athlete (AE) (*n* = 7) Youth elite athlete (YE) (*n* = 6) Youth nonelite athlete (YN) (*n* = 6)		↑ Microbial diversity in elite athletes >YN SCFA producers (*Clostridiales*, *Ruminococcaceae*, *Faecalibacterium*) were dominant in elite athletes	↑ ATP metabolism, many sugar transport systems, carbohydrate metabolism elite athletes

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TABLE 1. *(Continued)*

Animal Models

Article	Subjects	Exercise Stimulus	Diet	Test Groups	Main Outcomes	Gut Microbiota Findings	Metabolites/Metabolic Pathways Findings
Petersen *et al.* (31)	*N* = 33 Professional and amateur competitive cyclists		Professional cyclists Amateur cyclists	*Prevotella* correlated with exercising time; however, abundances reduced at 20 + h·wk^{-1} of exercise	↑ *M. smithii* in professional > amateur cyclists ↓ *Bacteroides* and ↑ *Akkermansia* abundances in all athletes	↑ Branched-chain amino acid, energy, and carbohydrate metabolism in professional cyclists ↑ Methane producing gene expression and methane metabolism in professional cyclists	
Grosicki *et al.* (32)	*N* = 12 Ultraendurance triathlon athletes	Samples collected pre- and post-race completion	Ultraendurance athletes	Exercise shifts microbial metabolism independent of microbial composition Resting-state microbial ecology determined by performance level and training history		↓ Free and secondary bile acids, SCFA (butyric and pivalic acid) ↑ Long-chain fatty acids (oleic and palmitoleic acids)	
Bycura *et al.* (33)	*N* = 56	8 wk of exercise cardiorespiratory exercise: 60-min sessions, 3× per week at 60%–90% max heart rate resistance training exercise: 60-min sessions, 3× per week at 70%–85% 1 repetition max, 3–6 sets of 6–12 reps	Cardiorespiratory exercise (CRE) (*n* = 28) resistance training exercise (RTE) (*n* = 28)	Baseline gut microbial composition may predict exercise gains	Initial changes to gut microbiome that were not sustained in CRE No significant gut microbiome changes in RTE		
O'Donovan *et al.* (34)	*N* = 37 elite Irish athletes		Categorized by dynamic (low, moderate, high) and static component (low, moderate, high)	Training load and type creates varying gut microbiome and metabolomic changes		*Cis*-aconitate, succinic acid, lactate, creatinine significantly differed between groups	
Gross *et al.* (35)	*N* = 7 healthy, physically active men and women	Run to exhaustion test	Crossover study Placebo (*n* = 7) Probiotic *Veillonella atypica* (*n* = 7)	*V. atypica* maintained performance ↓ Anaerobic capacity in placebo		Changes in lactate after exercise were similar in both groups Change in 14 metabolic expressions in *V. atypica*	
Gallè *et al.* (36)	*N* = 140 young, healthy Italian adults averaging 3006.2 MET·min^{-1}·wk^{-1} of physical activity			Diet and physical activity influence on gut microbiota biodiversity	↑ Prevalence of Bacillota and Bacteriodota ↑ *Streptococcus* and *Dorea* abundance in overweight/obese individuals ↑ *Ruminococcus* and *Oscillospira* in low Mediterranean diet adherence ↑ *Lachnobacterium* in low physical activity levels		
Hampton-Marcell *et al.* (37)	*N* = 13 Division I NCAA collegiate swimmers	Training log used to record data from peak training to the end of in-season taper	2016 season swimmers (*n* = 9) 2017 season swimmers (*n* = 7)[a]		↓ Microbial diversity with ↓ Training volume ↓ *Faecalibacterium* and *Coprococcus* with ↓ training volume		

[a] Three participants in both 2016 and 2017 seasons.

ABX, antibiotic; AC5KO, adenylyl cyclase 5 knockout; BMI, body mass index; DSS, dextran sodium sulfate; FMT, fecal microbiota transplantation; SPF, specific pathogen-free; Tph1, tryptophan hydroxylase 1; WHO, World Health Organization.

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placed in cages and animals engage in voluntary activity, whereas in FTR, animals are placed on a treadmill and then exercised at given speeds and grades, as in humans. Interestingly, studies have found differences in microbial diversity and metabolite concentrations between voluntary and forced exercise (6,9,12,14,28). Our lab has shown that 12 wk of VWR in male mice increased *F. prausnitzii* and other butyrate-producing bacteria independent of diet and reduced intestinal inflammation (11). This important finding was the first to show that exercise alone could manifest a unique gut microbiota profile. In an earlier study by Matsumoto *et al.*, 6-wk-old male Wistar rats exercised for 5 wk using VWR on controlled diets. PCR-TGGE and a cluster analysis of the banding profiles was used to observe gut microbiota changes as well as cecal contents to evaluate SCFA profiles (12). They found compositional changes to the gut microbiota including increases in butyrate-producing bacterium SM7/11 and T2–87, increased cecum size, and increased butyrate concentrations in exercised animals when compared to sedentary controls (12). Allen *et al.* sought to determine the difference between sedentary, VWR, and FTR in 6-wk-old male mice (6). Mice were controlled for the number of days of exposure to exercise in each group (n = 30) and on a standard diet but were not controlled precisely for exercise intensity or volume between the two different exercise modalities (6). VWR mice were provided with a wheel, which they had free access to and FTR mice ran 40 $\text{min}\cdot\text{d}^{-1}$ on a treadmill at a 5% grade for 8–12 $\text{m}\cdot\text{min}^{-1}$ for 6 wk. Findings showed that the community structures of the three treatment groups clustered separately from each other in both cecum and fecal samples ($P < 0.05$). Contrary to their hypothesis, α-diversity metrics indicated that VWR led to reduced bacterial richness compared with FTR or sedentary animals ($P < 0.05$). Furthermore, taxonomic evaluation revealed that both VWR and FTR altered many individual bacterial taxa. In particular, *Turicibacter* spp., which has been strongly associated with immune function and bowel disease, was significantly lower in VWR versus SED/FTR. (6). They concluded that VWR and FTR differentially alter the intestinal microbiome of mice. In a different study by Allen *et al.* (8), 6-wk-old male mice who exercised for 6 wk using VWR were used to test whether exercise-induced modifications in the gut microbiota can directly affect host responses to microbiota colonization and chemically induced colitis. To test this, "exercised microbiota" was transplanted to germ-free (GF) mice. They reported that microbial transplants from donor (exercised or control) mice led to differences in microbiota β-diversity, metabolite profiles, colon inflammation, and body mass in recipient mice 5 wk after colonization. Importantly, they demonstrated that colonization of mice with a gut microbiota from exercise-trained mice led to an attenuated response to chemical-induced colitis, evidenced by reduced colon shortening, attenuated mucus depletion and augmented expression of cytokines (8). Specifically, mice colonized with an "exercised gut microbiota" had significantly higher expression of cytokines and transcription factors known to be involved in tissue regenerative responses to colitis, including TGFβ, FoxP3, and IL-22. Furthermore, histology showed a significant attenuation of goblet cell mucus depletion in these mice. Taken together, this evidence nicely shows increased markers, like TGFβ and IL-22, and reductions in mucus depletion could contribute to improvements in intestinal barrier function and mitigate colitis symptomatology supporting the conclusion that exercise-induced modifications in the gut microbiota can mediate host–microbial interactions with potentially beneficial outcomes for the host.

Mika *et al.* (14) revealed that 6 wk of VWR in the juvenile stage (24 d postnatal) rather than the adult phase mice (70 d postnatal) generated a more robust and substantial change to the gut microbiota (14). Furthermore, the juvenile mice had greater SCFA production, increased Bacteroidota, and reduced Bacillota abundances, which are associated with a lean phenotype. The increase in SCFA was shown to reduce fat accumulation in adipose tissue and increase energy expenditure. Microbes associated with a lean phenotype, *Bifidobacteria* spp. and *Methanosphaera* spp., increased, but only in the juvenile group. Moreover, the lean phenotype and gut microbiota changes in the juvenile group persisted 25 d after exercise. It was concluded that the diversity of juvenile mice gut microbiota was more plastic and sensitive to gut microbiota changes than the adults.

It should be noted that not all studies report changes in gut microbiota following exercise. Lamoureux *et al.* (13) examined forty-two 6- to 10-wk-old C57BL/6 mice (11 male, 31 female) who underwent 8 wk of FTR (40 $\text{min}\cdot\text{d}^{-1}$ for 5 $\text{d}\cdot\text{wk}^{-1}$ progressing from 15 $\text{m}\cdot\text{min}^{-1}$ to 20 $\text{m}\cdot\text{min}^{-1}$) or VWR (free access), or who remained sedentary. They reported that using traditional microbial profiles comparing operational taxonomic units (OTUs) in samples ($P > 0.1$) and multivariate analysis of beta diversity via Adonis testing ($P > 0.1$) did not identify significantly altered taxonomic profiles in the voluntary or forced exercise group compared to the sedentary controls. However, a random forests machine learning model, which considers the relationships between bacteria in a community, classified voluntary exercisers and nonexercisers with 97% accuracy at 8 wk. The top bacteria used by the model identified known taxa (*Bacteroides*, S24–7, and *Lactobacillus*) and novel taxa (*Rikenellaceae* and *Lachnospiraceae*) associated with exercise (13).

Many of these early studies (6,8,11–13) only report high levels of taxonomic resolution (*e.g.*, phyla, family, and genus, which are broad and contain many different types of gut microbes with a variety of functions). However, examination to the species and strain level resolution can contribute immensely to the field and identify gut microbes that may be novel and particularly impactful for human health. In support of this, our 2020 study, using long read technology (Oxford Nanopore MinION) showed that exercise can produce more robust changes to the gut microbiota than genotype as well as identified bacterial strains associated with exercise only (*Muribaculum Intestinale YL27*) and sedentarism only (*Turicibacter sanguinis*) (38). As such, there may be a need to incorporate long reads into gut microbiota analysis to fully elucidate species and strain level changes that result from exercise interventions. (see Fig. 1 for summary of beneficial by-products of gut microbes on performance).

Exercise trained compared to sedentary human subjects

It has been noted that the gut microbiome of a sedentary individual differs from an active individual. Langsetmo *et al.* (24), Bai *et al.* (25), and Bressa, *et al.* (10) analyzed data based on the individuals' present physical activity levels at the time of study and did not ask the participants, men, children, and women

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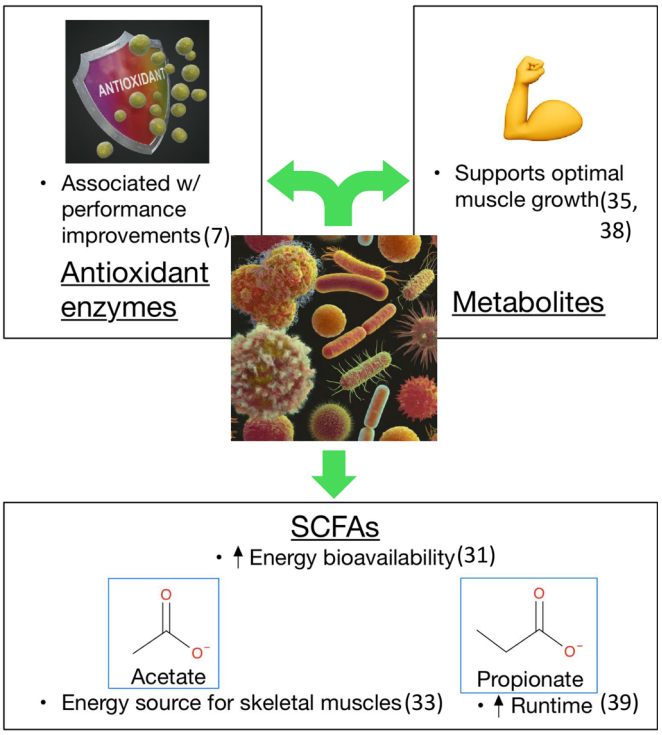

Figure 1. Beneficial by-products of gut microbes on exercise performance. Gut microbes produce SCFA, antioxidants, and metabolites as by-products, which aid in improvements in exercise performance. Antioxidants image generated by Gemini (https://gemini.google.com/app/0a8be09e5ab5cd99). Microbes image generated by Gemini (https://gemini.google.com/app/0a8be09e5ab5cd99).

respectively, to change their activity level. Langsetmo *et al.* found that men 65+ yr old, who exercised as tracked as step activity and total energy expenditure, had increases in abundances of *Faecalibacterium* and *Lachnospira* spp. (24). Bai *et al.* used data from the American Gut Project and revealed a positive association between exercise and Bacillota, specifically, *Clostridiales*, *Lachnospiraceae*, and *Erysipelotrichaceae* in children ages 7–18 yr (25). Bressa *et al.* (10) found higher abundance of health-promoting bacterial species in active women (using accelerometer data), including *F. prausnitzii*, *Roseburia hominis*, and *A. muciniphila*. They also reported that with regard to muscular mass index (MMI) and appendicular muscular mass index (AppMMI), *Faecalibacterium* was positively associated with MMI and AppMI, and *Coprococcus* and *Lachnospiraceae* unclassified 1 were positively correlated with AppMMI suggesting a link between gut microbiota and muscle mass (10). More will be discussed about this link below.

Training studies focused on gut microbiota compositional and functional changes have also revealed that exercise is beneficial (26–28). Morita *et al.* revealed an increase of *Bacteroides* spp. with aerobic exercise, 60 min at a ≥3 METs, in a 12-week long nonrandomized study with elderly women (65+ yr old) (26). Munukka *et al.* (27) matched frequency and intensity in a study of sedentary, untrained Finnish women on their normal *ad libitum* diet who were cycling for 6 wk with gradual increases in intensity every 2 wk. Results show that β-diversity increased from 0.449 to 0.465 (P = 0.014) in response to exercise training, whereas phyla and family levels did not change. In addition, exercise training resulted in a decrease of gram-negative phylum Proteobacteria (P = 0.039) and an increase in Verrucomicrobia (P = 0.038). Specifically, exercise increased Verrucomicrobiaceae (P = 0.038). Further, the genus and *Akkermansia* (P = 0.038) increased after exercise training.

Allen *et al.* (28) explored the impact of 6 wk of endurance exercise on the composition, functional capacity, and metabolic output of the gut microbiota in lean and obese adults with multiple-day dietary controls before outcome variable collection. Thirty-two lean (n = 18 (9 female)) and obese (n = 14 (11 female)), previously sedentary subjects participated in 6 wk of supervised, endurance-based exercise training (3 $d \cdot wk^{-1}$) that progressed from 30 to 60 $min \cdot d^{-1}$ and from moderate (60% of HR reserve) to vigorous intensity (75% HR reserve). Subsequently, participants returned to a sedentary lifestyle activity for a 6-wk washout period. Fecal samples were collected before and after 6 wk of exercise, as well as after the sedentary washout period. β-Diversity analysis revealed that exercise-induced alterations of the gut microbiota were dependent on obesity status. Individuals with the lean phenotype, when compared to those with an obese phenotype, had a significantly greater production of SCFA supported by the increased concentration of the enzyme butyryl-CoA transferase responsible for generating acetyl-CoA and butyrate. Lastly, exercise-induced changes in the microbiota were largely reversed once exercise training ceased, suggesting that the exercise stimulus is required for sustaining beneficial microbes and SCFA (28).

In summary, there is evidence to demonstrate a beneficial gut microbiota in already exercising individuals as well as sedentary individuals who begin to exercise. Exercise-induced microbial changes in human studies were observed across all ages and sex. Furthermore, like animal studies, human studies show increased SCFA-producing bacteria. Finally, the exercise regimen must persist to maintain exercise-induced gut microbiota changes or else changes resort to baseline, suggesting that changes to the gut microbiota to exercise are a training response.

Highly trained human subjects

Elite athletes have a compositionally different gut microbiota than sedentary and recreationally active individuals who can be advantageous to performance (1). Many studies have similarly shown that elite athletes have increased alpha diversity (29,39) and greater Bacillota/Bacteroidota ratios (30). In addition, there is a greater selection for SCFA producers. Compositionally, exercise stress has been shown to increase the abundance of *Akkermansia* (1,5,9), *Ruminococcus gnavus*, *Butyrivibrio*, *Oscillospira*, and *Coprococcus* while decreasing *Turicibacter* (5). Many of these microbes have functions that may enhance health. For example, besides being a SCFA producer, *Akkermansia* also plays a role in mucin degradation to reduce inflammation and strengthen tight barrier junctions (40).

Han *et al.* (30) and Barton *et al.* (9) examined professional athletes and showed not only compositional changes to the gut microbiota but also metabolic changes. Han *et al.* (30) showed that there was greater carbohydrate and ATP metabolism, oxidative stress tolerance, and increased variety of sugar transports found in the guts of elite rowers. Although Barton *et al.* (9) revealed that chronic exercise in professional rugby players promotes gut microbes capable of organic cofactor and antibiotic synthesis, increased expression of carbohydrate degradation, and secondary metabolite metabolism. Additionally, various protein degradation products such as trimethylamine N-oxide, carnitines, trimethylamine, 3-hydroxy-isovaleric acid,

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and phenylacetylglutamine were found in greater abundance within athletes (9).

A pilot study conducted by Petersen *et al.* used metagenomic whole genome shotgun and metatranscriptomic (RNA-Seq) sequencing to show what organisms are both present and active in the gut microbiomes of professional and amateur-level competitive cyclists. They reported that no significant correlations could be found between taxonomic clustering and being either a professional or amateur-level cyclist; however, high abundance of the genus *Prevotella* (≥2.5%) was significantly correlated with time reported exercising during an average week (31). In addition, cyclists who trained greater than 11 $h \cdot wk^{-1}$ had high levels of *Prevotella* and *Methanobrevibacter smithii*. The increased abundance of *Prevotella* was correlated with several amino acid and carbohydrate metabolism pathways. Specifically, correlations between KEGG pathways and the abundance of *Prevotella* showed that *Prevotella* was positively correlated ($\rho > 0.5$, $P < 0.01$) to several amino acid metabolism pathways, including lysine biosynthesis, alanine, aspartate and glutamate metabolism, and D-glutamine and D-glutamate metabolism. *Prevotella* was also significantly correlated to multiple pathways involved in drug metabolism, carbohydrate metabolism, and metabolism of cofactors and vitamins, including vitamin B_6 metabolism. Further analysis of the metatranscriptome revealed an increased abundance of *M. smithii* transcripts in several professional cyclists, and this archaeon had upregulation of genes involved in the production of methane. Furthermore, when methane metabolism was upregulated, there was similar upregulation of energy and carbohydrate metabolism pathways. Taken together, this study suggests that *Prevotella* and *M. smithii* may be important microbes for exercise given the known benefits of amino acids for muscle growth and function, as well at their link cofactors and vitamins that may be crucial for energy metabolism. Larger clinical trials may be needed to elucidate this as well as to uncover additional microbes and their associated pathways and functions that are meaningful to exercise.

Grosicki *et al.* (32) characterized acute changes in the fecal microbiome and metabolome in 12 triathletes following participation in an ultraendurance triathlon (3.9 km swim, 180.2 km bike, 42.2 km run). Stool samples were collected within 48 h to the start of the triathlon and the first bowel movement postrace. Although intraindividual and interindividual diversity of bacterial species and individual bacterial taxa were unaltered following race completion ($P > 0.05$), significant reductions ($P < 0.05$) in free and secondary bile acids (deoxycholic acid (DCA), 12-keto-lithocholic acid (12-ketoLCA)) and short-chain fatty acids (butyric and pivalic acids) were seen. In addition, significant increases ($P < 0.05$) in long-chain fatty acids (oleic and palmitoleic acids) were observed. They also reported an association between higher abundances of *A. muciniphila* and shorter race times; however, interestingly, a greater abundance of *M. smithii* was associated with slower triathlon finishing times, which was hypothesized to have aided in sustaining the participants through the longer race. Inverse correlations were found between race time and free bile acids, palmitic acids, saturated fatty acids, linoleic, and oleic acids. Meanwhile, isovaleric acid concentrations had a positive correlation with race times. Of note, an exploratory aspect of Grosicki *et al.* (32), which examined years of training and bacterial species, revealed that athletes who had trained for a longer duration (*i.e.*, >12 yr) possessed a greater abundance of *Ruminococcus lactaris* and *R. gnavus*, as well as *Bacteroides* spp. (*Bacteroides thetaiotaomicron*, *Bacteroides salyersiae*, *Bacteroides plebeius*, *Bacteroides faecis*, and *Bacteroides coprocola*) ($P < 0.05$). Meanwhile, athletes who had trained for less than 12 yr possessed a greater abundance of *Ruminococcus torques* and *Roseburia*_cp_CAG_471 ($P < 0.05$).

In summary, evaluation of elite athletes, length of training, and gut microbiota composition deserves more attention as well as more mechanistic approaches to understand why there may be difference in composition based on years of training. Furthermore, the field is in dire need of time course studies to see when these changes may occur, how they are sustained, and what, in addition to detraining, may alter or reverse the impacts exercise has on the gut microbes.

Impacts of exercise intensity and modality on gut microbiota in human subjects

The mode and intensity of exercise and the gut microbiota have received far less attention experimentally. To date, studies like Grosicki *et al.* (32), Liang *et al.* (29), Gallè *et al.* (36), and Hampton-Marcell *et al.* (37) show that years of training, hours per week training, METs per week, or time within a season (as measured by intensity and time training) all modulate the gut microbiota in a dose–response manner, meaning that the more exercise, the more the microbes change in response to the exercise stimulus (details of these studies are found in Table 1).

To examine mode, Bycura *et al.* compared aerobic exercise to anaerobic resistance training exercise (RTE). Twenty-eight subjects (21 F; aged 18–26 yr) were recruited for the aerobic portion of the study and 28 subjects (17 F; aged 18–33 yr) were recruited for resistance training portion of study. Fecal samples for gut microbiome profiling were collected twice weekly during the preintervention phase (3 wk), intervention phase (8 wk), and postintervention phase (3 wk); profiled using 16S rRNA gene sequencing; and analyzed with QIIME 2. Aerobic training resulted in initial changes, largest changes between weeks 2 and 3 of training, but were not sustained through or after the intervention period. Furthermore, the group noted an increase in RER and time spent on the treadmill, which was associated with *Prevotella*. However, resistance training resulted in no detectable changes to the gut microbiota. (33).

O'Donovan *et al.* (34) set out to determine if differences existed in the gut microbiome and metabolome of elite-level athletes participating in different categories of sports. To gain an understanding of the impact of dynamic versus static components of exercise, each sport was classified into broader sports classification groups by letters A, B, and C (low, moderate, or high $\dot{V}O_{2max}$) and numbers 1, 2, and 3 (low, moderate, and high maximal voluntary contraction). They showed that *Streptococcus suis*, *Clostridium bolteae*, *Lactobacillus phage LfeInf*, and *Anaerostipes hadrus* were found to be associated with those groups with a moderate dynamic component (B1 and B2), which includes sports such as fencing. *Bifidobacterium animalis*, *Lactobacillus acidophilus*, *Prevotella intermedia*, and *F. prausnitzii* were found to be associated with the C1 (high dynamic and low static components, including sports such as field hockey), whereas *Bacteroides caccae* was found to be associated with the C3 (high dynamic and static components, including sports such as rowing). Samples from the group C3 were found to have 4.5 times greater relative abundance of *B. caccae* than other groups,

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whereas *Anaerostipes hadrus* was present at 3 times greater relative abundance in the group B relative to other groups. *F. prausnitzii* was present at 1.5 times greater relative abundance in samples from group C1 in comparison to other groups. Interestingly, no species were identified as being associated with A3 (low dynamic and high static components, including sports such as judo) or C2 (high dynamic and moderate static components, including sports such as swimming).

Overall, different forms of exercise generate different gut microbial changes. Additionally, across the various aerobic sports, there are different microbial and metabolic changes that perhaps evolve to better sustain the host's demands dependent on sport and exercise intensity. This area is one that requires more research so that sport-specific gut microbiotas may be profiled and potentially used to help enhance performance.

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Bidirectional Evidence of the Gut Microbiota and Exercise Conducted in Mice

As summarized above, many animal and human studies examine how exercise alters the gut microbiome; however, there are far fewer studies that have been conducted that test how a depleted gut microbiome impacts exercise performance. Through studies using germ-free and gnotobiotic mice (mice that have known microbes transplanted into their gut), researchers have observed increased performance in mice with gut microbiota compared to those animals that lack gut microbes (7,16,18,19). Hsu *et al.* (7) analyzed the role of the gut microbiota on exhaustive swim test time in 12-wk-old male, specific pathogen-free (SPF) (n = 8), GF (n = 8), and *Bacteroides fragilis* gnotobiotic C57BL/6JNarl (n = 8) mice. Results showed that germ-free mice had the shortest swim time, followed by the *B. fragilis* gnotobiotic mice, whereas the SPF mice had the greatest swim time to exhaustion. Additionally, it was reported that GF mice had lower antioxidant enzyme activity (*e.g.*, serum and hepatic glutathione peroxidase, and serum catalase). Liver catalase demonstrated no differences between any groups. Interestingly, *B. fragilis* gnotobiotic mice had significantly lower serum (but not liver) superoxide dismutase activity compared to both GF and SPF, with no differences between GF and SPF mice. Overall, the results suggest that gut microbes are needed for exercise capacity and their potential action linked with the antioxidant enzyme system (7). In addition, gut microbiota may be differentially linked to serum and liver antioxidant enzyme activity; however, this requires experimental exploration (Fig. 2).

Huang *et al.* (16) used 6-wk-old male C57BL/6JNarl gnotobiotic mice inoculated with *Eubacterium rectale*, *Lactobacillus plantarum*, and *Clostridium coccoides* (n = 6–8 mice/each microbe) to evaluate these individual microbe's effects on exercise performance. Inoculation began 1 wk before the exercise testing/training and then was performed weekly during the exercise for a total of 6 wk. Endurance performance was measured using exhaustive swim tests, and all the mice in the four groups were subjected to swim training 10 min·d^{-1}, 5 d·wk^{-1} between weeks 3 and 7. Fecal samples from the four individual groups were collected at weeks 1, 3, 5, and 7 before and after oral inoculation. Mice were provided *ad libitum* access to standard rodent diet and water. First, Huang *et al.*'s findings supported the earlier report by Hsu *et al.* (7) that GF mice had significantly lower time to exhaustion compared to SPF mice. New to be reported was that GF mice were capable of training adaptations demonstrating significantly increased time

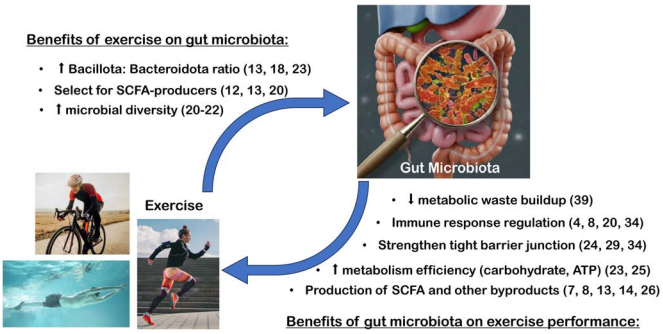

Figure 2. Bidirectionality of gut microbiota and exercise. Exercise has shown beneficial effects on gut microbiota composition and selection for microbes that aid in overall health and sport-specific demands. Gut microbiota image generated by Gemini (https://gemini.google.com/app/6e080e5341d77af3). Exercise photos from MS PowerPoint stock photos.

to exhaustion after the 5 wk of swim training. *L. plantarum* and *C. coccoides* did not colonize stably in rodent intestines; however, *E. rectale* showed a significant incremental increase over time. *E. rectale* engendered mice had significantly increased time to exhaustion for postsupplementation and posttraining compared to the other two microbes. In addition, *L. plantarum* TWK10 and *C. coccoides* were found to significantly increase basal metabolic rate and to alter the body compositions, although no exercise capacity benefit was found in these gnotobiotic mice. In assessing markers of exercise-induced injury *E. rectale* and *L. plantarum* all had lower serum creatine kinase levels, but there were no differences in lactate dehydrogenase. In addition, serum lactate levels were significantly lower in *E. rectale* and *C. coccoides* compared to GF and *L. plantarum*. Serum glucose levels showed significant differences, with the *E. rectale* and *C. coccoides* groups having significantly higher values than the GF group ($P < 0.05$) and no significant difference found between the GF and *L. plantarum* groups. Finally, *E. rectale* group showing significantly higher muscle tissue content of GLUT4 than the GF and *L. plantarum* groups. Although this study was exciting, there appears to be no microbe consistently linked to all the study outcomes suggesting that it is likely that there are groups, rather than individual microbes, responsible for exercise performance and training adaptations, which desperately need research attention.

It was recently reported that an intact gut microbiota is important for maximal muscle development. Valentino *et al.* (17) investigated hypertrophy of the soleus and plantaris muscle in female mice with an intact gut microbiota and an antibiotic-induced dysbiosis. Forty-two 4-month-old female mice were randomly assigned to one of four groups: untreated (U) or antibiotic-treated (T) nonrunning controls (C) or progressive weighted wheel (T) running. Despite a similar training stimulus per session (12.35 ± 2.06 vs 11.09 ± 1.76 km·wk^{-1}, respectively) and total (778.9 ± 130.5 vs 703.8 ± 112.9 km, PU vs PT) between the groups, PT showed less hypertrophy of soleus type 1 and 2a fibers and plantaris type 2b/x fibers compared to PU. Furthermore, the higher satellite cell and myonuclei abundance of PU plantaris muscle after PoWeR was not observed in PT. Finally, the fiber-type shift of PU plantaris muscle to a more oxidative type 2a fiber composition following PoWeR was blunted in PT. It was hypothesized that the antibiotic-treated (ABX) group lacked a necessary gut microbiota-produced metabolite that optimally supports muscle growth, which has not been identified yet. Moreover, exercise induced a shift from fast

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to slow twitch muscle fibers in the healthy controls, which was not observed in the plantaris muscle of the antibiotic-treated mice. In conclusion, lacking an intact gut microbiota causes the inability to maximally adapt, via hypertrophy, to exercise training (17).

A study by Okamoto *et al.* examined the effects of exercise performance on 10-wk-old male mice administered either multiple ABX or a low microbiome (LMC) or high-microbiome accessible carbohydrate (HMC) diet, before endurance testing on a treadmill. Initial results, like other studies, showed that antibiotic-treated mice had significantly reduced exercise capacity as measured by exhaustive treadmill run time. Okamoto *et al.* also reported that treadmill running time was significantly lower in the LMC than the HMC diet group, as was plasma acetate and propionate concentrations. Because acetate was the most abundant SCFA, experiments to continuously infuse acetate were conducted in antibiotic-treated mice and showed that this continuous infusion ameliorated reduced running times compared to saline and butyrate. Furthermore, fecal transplant and administration of fermentable fiber to the LMC group normalized exercise tolerance. Conclusions based on this study were that acetate generated in the intestine by microbes is an important energy substrate during endurance exercise (18).

Given that research on exercise and changes to gut microbiota usually shows changes to butyrate producers, enzymes, and concentration, whereas gnotobiotic studies demonstrate that propionate (described below) and acetate (18) are linked to enhanced exercise capacity, this raises questions around which SCFA may be most beneficial. One could state that, taken together, these contradictory results suggest that all the short-chain fatty acids may provide some benefit to the host with regard to exercise performance, but what that benefit is may be unclear given the lack of studies in this area.

Dowden *et al.* (19) investigated exercise performance in 3- to 6-month C57BL/6J male mice before and after 5 wk of exercise training as well as after 1 wk of antibiotic administration to determine if ABX interferes with adaptations to exercise training. Performance was assessed by a graded exhaustive treadmill test set to a 10% grade, which remained consistent throughout the test. The treadmill began at 4 $m \cdot min^{-1}$, and the speed incrementally increased 2 $m \cdot min^{-1}$ every 2 min until the mice reached exhaustion. Exercise training was FTR for 1 $h \cdot d^{-1}$ for 5 $d \cdot wk^{-1}$ for 5 wk where mice progressed to 70% maximum speed during the first 2 wk and remained at 70% maximum speed for the remaining 3 wk. Once again, study findings were consistent with another (18) showing that ABX caused a significant reduction in the body, liver, and skeletal muscles. Exercise training revealed 60% and 63% increase in run distance and work to exhaustion, respectively, compared to baseline. However, after the consumption of the antibiotic cocktail, these improvements in performance were reversed to baseline. Furthermore, increases induced by exercise training in hindlimb blood flow, nitric oxide concentrations, mitochondrial oxidative phosphorylation (complex IV and citrate synthase), and mitochondrial biogenesis (SIRT1, AMPK, and PGC-1α) were all abolished upon the introduction of ABX (19). The major findings of this investigation involved two mechanisms that mediate training-induced enhanced exercise performance (blood flow and mitochondrial function) and those mechanisms mediating loss of training-induced enhanced exercise capacity with the deletion of the gut microbiota. This demonstrated that the gut microbiota plays an important role in mediating the increases in exercise capacity after exercise training, which, in turn, is mediated by increases in aerobic capacity, increases in hindlimb blood flow, and mitochondrial function.

To further support the role of the gut microbiota in mediating exercise capacity, Nay *et al.* (20) assessed the effects of ABX (21 d) and natural reseeding (10 d) on the gut microbiota and performance. Natural reseeding is the process of using a mice's inherent nature, coprophagy, to reintroduce gut microbes. In this study, natural reseeding would be to provide control group feces to the ABX-treated mice. There was no exercise training between the exhaustive run tests, and mice were allowed *ad libitum* food and water. Results also showed that exercise, as measured by running distance, in gut microbiota depleted mice, was decreased, as well as the extensor digitorum longus muscle fatigue index in an *ex vivo* contractile test. Importantly, the muscle endurance capacity was efficiently normalized by natural reseeding. Additional findings showed that the absence of the gut microbiota was associated with an increase in insulin sensitivity and reduction in serum glucose hypothesized to be due to a lack of SCFA producers and production of butyrate. In support of this, Nay *et al.* showed that ileum gene expression of short fatty acid chain and glucose transporters G protein-coupled receptor 41 and sodium-glucose cotransporter 1 and muscle glycogen level, paralleled the muscle endurance changes observed after treatment with ABX for 21 d and reseeding (20). Like the Dowden *et al.* (19) and Valentino *et al.*, (17) studies, Nay *et al.*'s study results suggest a gut microbiota–skeletal muscle axis, which supports the hypothesis that gut bacteria are required for host optimal skeletal muscle function (17,19,20).

Hutchinson *et al.* (21) sought to determine the role of gut microbes in adaptations to exercise training and performance in male and female C57Bl/6 mice (n = 56) provided access to VWR for 6 wk. After the intervention, treadmill endurance and glucose tolerance were assessed, and gastrocnemius and soleus tissues were harvested and analyzed for citrate synthase enzyme activity and expression of exercise training–sensitive genes. Findings showed that ABX treatment and GF status resulted in VWR volumes ~22% and 26% lower than controls, respectively. However, ABX did not hinder performance improvements induced by endurance training. The authors reported that, although ABX-treated mice performed numerically lower (~5%–13% reduction dependent on measure) on the treadmill test post-training, this difference was not statistically significant. Importantly it was noted that GF VWR mice exhibited significant improvements in time to fatigue and maximal running speed when compared to GF sedentary mice. VWR increased gastrocnemius citrate synthase enzyme activity in all groups, and ABX and GF status did not reduce the VWR effect. VWR also increased muscle expression of PGC1-α, but this was not affected by ABX treatment. Hutchinson *et al.*, hypothesized that it was not the sole absence of gut microbes that negatively impacted performance, but rather the lack of microbes since birth that inhibits the development of exercise-induced adaptations. Contrary to much of the literature, the antibiotic group demonstrated no attenuation of skeletal muscle adaptation indicating that the gut microbiota may not be needed to obtain training adaptations, but may indicate that the presence of gut microbes may play a role in exercise motivation (21).

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Dohnalová *et al.* (22) conducted a series of experiments to establish a link between exercise motivation and the gut microbiota. They reported the discovery of a gut–brain connection in mice that enhances exercise performance by augmenting dopamine signaling during physical activity. Specifically, they showed that microbiome-dependent production of endocannabinoid metabolites in the gut stimulates the activity of transient receptor potential vanilloid 1 or TRPV1-expressing sensory neurons and thereby elevates dopamine levels in the ventral striatum during exercise. Stimulation of this pathway improves running performance, whereas microbiome depletion, peripheral endocannabinoid receptor inhibition, ablation of spinal afferent neurons, or dopamine blockade abrogates exercise capacity. These findings indicate that the rewarding properties of exercise are influenced by gut-derived interoceptive circuits and provide a microbiome-dependent explanation for interindividual variability in exercise performance. Because this review is not focused on exercise motivation, the reader is encouraged to consult the reference to explore other studies investigating this exciting area of exercise–motivation–gut microbiota link.

In additional efforts to understand skeletal muscle function and the gut microbiota, Lahiri *et al.* (23) set out to elucidate the functional interactions between the gut microbiota and regulation of skeletal muscle mass and function using GF and SPF C57BL/6J 6- to 8-wk-old male mice. This comprehensive design examined skeletal muscle mass and differences in expression of genes encoding for different cell signaling molecules that regulate muscle mass. Further, detailed metabolic characterization was performed using NMR spectrometric analyses of the skeletal muscle, liver, and serum samples. In addition, the influence of the gut microbiota on neuromuscular junctions was also examined by analyzing expression of genes encoding neuromuscular junction proteins and testing mouse muscle strength in the grip strength. Finally, the gut microbiota's role in muscle mass and function was further analyzed by transplanting GF mice with the gut microbiota of PF mice or by treating them with microbial metabolites (SCFAs). Results showed that GF mice lacking a gut microbiota had reduced skeletal muscle weight compared to SPF mice and that when GF mice were transplanted SPF gut microbiota, this restored muscle mass. Further, GF mice had significantly reduced expression of the myosin heavy-chain genes MyHCIIa, MyHCIIb, and MyHCIIx in the tibialis anterior. Atrophy genes were increased in GF mice, whereas muscle differentiation genes were reduced. Coupled with this, GF mice showed reductions in muscular strength. Mice fed with SCFA were found to have partial restoration of skeletal muscles, whereas GF mice inoculated with the gut microbiota of the pathogen-free mice demonstrated an increase in skeletal muscle mass and oxidative metabolic capacity and decreased atrophy biomarkers (23). This expansive study concluded that activation of AMPK in GF mouse muscle suggests that the AMPK–FoxO3–Atrogin cascade could be one possible signaling pathway involved in atrophy or that observations of increased corticosterone concentrations in serum, induction of Klf15 gene expression, and activation of the BCAA pathway indicated a possible HPA-glucocorticoid–driven atrophy of skeletal muscle mass in GF mice. Taken together this suggests that multiple pathways may be involved in regulation of skeletal muscle mass and that the gut microbiota and its microbial derived metabolites may be linked to these pathways. Finally, because muscles of different metabolic subtypes showed similar findings, these pathways may be linked to all types of skeletal muscle. However, much more research is needed in this area. What is clear from the studies summarized is that 1) there is a link, albeit complicated, between gut microbiota and skeletal muscle mass and function, and 2) additional research is needed to fully elucidate the mechanistic links.

To date, not many human studies have been conducted on antibiotic use or specific microbial supplementation on performance. One such study that has examined this used a translational model first examining the gut microbiota of marathon runners (n = 15, running the Boston marathon, sex not noted in the study) compared to healthy controls (n = 10) (15). 16S rDNA sequencing on samples collected daily up to 1 wk before and 1 wk after marathon day showed that *Veillonella atypica* was found in the gut of the marathon runners, which has been shown to convert lactate to acetate and propionate. The second set of experiments to understand what or how *V. atypica* physiologically works utilized an AB/BA crossover mouse experiment spanning 2 wk consisting of a control group (*Lactobacillus bulgaricus* gavage, n = 16) and a treatment group (*V. atypica*, n = 16) with a treatment/control crossover happening between weeks. *L. bulgaricus* was chosen as a control due its inability to catabolize lactate, mimicking bacterial load without impacting lactate metabolism. Results showed that the *V. atypica* mice ran 13% longer than the control group, and that the metabolic pathway that *Veillonella* species utilize for lactate metabolism was also enriched. Finally, to confirm whether systemic lactate can cross the epithelial barrier into the gut lumen, they performed tail-vein injections of $^{13}C_3$ sodium lactate into mice colonized with either *V. atypica* or *L. bulgaricus* and identified this label in the gut lumen. Further testing demonstrated that *V. atypica* was unable to ferment carbohydrates and that lactate was the sole carbon source using the methylmalonyl-CoA pathway. Mice with the FMT of *V. atypica* also had lower levels of inflammatory cytokines but no changes to muscle physiology due to the authors' notes as a lack of change in concentration of the glucose transporter, GLUT4. Based on their extensive studies and database querying, Scheiman *et al.* concluded that *V. atypica* may alone possess the full pathway to convert lactate into propionate and that the propionate alone was responsible for the increased runtime (15).

To test this further in humans, Gross *et al.* studied the impact of *V. atypica* supplementation on exhaustion times and anaerobic capacity. The randomized, double-blind, crossover study supplemented healthy, human participants for 14 d with a placebo or 1×10^{10} CFU doses of the probiotic *V. atypica* FB0054. Exhaustion times and anaerobic capacity were not significantly different between groups. Furthermore, no changes in lactate, hemodynamics, or bacterial community changes were observed (35).

In summary, most of the models show that both ABX and GF mice show reductions in exercise performance/capacity, and maximal muscle development and muscular function. However, this is only the beginning of these studies as future work, specifically what microbes or groups of microbes contribute to exercise performance or are critical for adaptation to exercise. Finally, as documented throughout this review, there is a severe lack of studies that include female animals or humans. Given the established sex differences in hormones, gut microbiota

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composition, and exercise capacity, this should be aggressively pursued in future work.

CONCLUSIONS

Exercise can alter the gut microbiota composition. As seen in various studies, alpha diversity and Bacillota abundance increases (25,28,40), whereas Bacteroidota decreases with exercise (1,9,30). Conversely, the reverse holds true for sedentary individuals (16). *Akkermansia*, *Lactobacillus*, *Bifidobacterium*, and other SCFA producers commonly have a positive correlation with exercise (1,9–11,27,30,31,34,35). These results are common to voluntary exercise; however, forced exercise generates results seen in chronic stress (6,17,28).

Evidence points to the gut microbiota adapting to sport-specific demands in response to exercise (16,18,19). The changes in the gut microbiota generated an accumulation of various metabolites that provided the host with performance advantages (16,18–22). Athletes typically have improved efficacy of carbohydrate metabolism, tolerance to oxidative stress and insulin sensitivity, muscle tissue repair, and energy harvest (9,17,31,33,40).

Finally, results from ABX and GF mouse models demonstrate a need for the gut microbiota to be intact for exercise performance and various aspects of maintenance of exercise training but perhaps not for adapting to exercise training (7,16–23).

GAPS AND FUTURE DIRECTIONS

Examination of the relationship between exercise and gut microbiota with performance is still in its infancy stage. Current research is examining how supplementation can improve the gut microbiota to indirectly lead to an enhancement in performance. It is important for future research to discover how we can use diet and supplements as a preventative measure to minimize the damages caused by the stress of exercise to maintain a healthy gut microbiota that optimizes performance. Current treatment options have tried to directly alter the gut composition to create a more optimized and healthier gut microbiota through probiotics, prebiotics, or synbiotics. However, future research should also include preventative measures to directly mitigate exercise-induced damages without deliberately changing the gut microbiota.

Studies examining lower levels of taxonomic resolution may uncover exercise-induced species and strains that may be critical for exercise that are currently not seen in experiments that only report to the family or genus. Strain-level determination can also be important because it demonstrates the functional genes in use rather than describing the genes present. This lack of specificity at the species and strain level makes translating research findings into clinical usage more difficult (41).

Future research should focus on discovering causative agents correlated with performance and understanding the physiological mechanisms rather than finding associations. To discover causation, Koch's postulate must be followed. Distinctly, Koch's postulate states that the causative agent of disease must be isolated, and the disease must be replicated when inoculating causative agents into healthy animals (41). Fei and Zhao followed Koch's postulate and were able to determine that *Enterobacter cloacae* B29 can single-handedly induce obesity promoted by a high-fat diet using gnotobiotic mice (42).

Another method of analysis that should be used is performing guild-based analyses with microbiome-wide association studies (MWAS). MWAS allows researchers to observe microbes in guilds, groups of microbes that work together and have co-abundant behavior. Typically, microorganisms from differing taxa have been observed to work with one another (43). Thereby, guild-based analysis would help define the structural and functional relationships and provide the emergent function rather than examining individual functions of various parts when studying specific microbes (43). Future studies should use a reference-free approach to reveal new knowledge rather than using a taxon-based analysis that is dependent on current knowledge. Most studies focus on first identifying and naming gut microbes using a database. Furthermore, scientists typically neglect microbes that are not identified due to the lack of information, thus creating a bias toward previously known knowledge. Not only does eliminating unknown microbes reduce the number of new microbes that could be added to the database, but it also would skew the data to be more misleading. With a reference-free approach, one is more likely to identify a new microorganism not previously published and produce more accurate results (44). Finally, studies focusing on microbial function rather than characterization are urgently needed. In support of this, Dowden *et al.* (38) examined species in the adenylyl cyclase-knockout model of healthful aging and revealed that, despite the improved exercise performance and metabolic phenotype of this mouse, microbes known to be pathogenic were high and those believed to be beneficial were low in the gut of these animals (38). This suggests that microbes may have more than one function given the metabolic state of the host, which could either promote a harmful or helpful behavior.

In conclusion, future research must fully translate animal findings to humans and use both sexes. Much of the work done thus far on microbiota depletion to determine the impacts on exercise has been done in animals. With the high rates of antibiotic prescription in humans, deciphering their impact of human exercise performance is paramount.

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